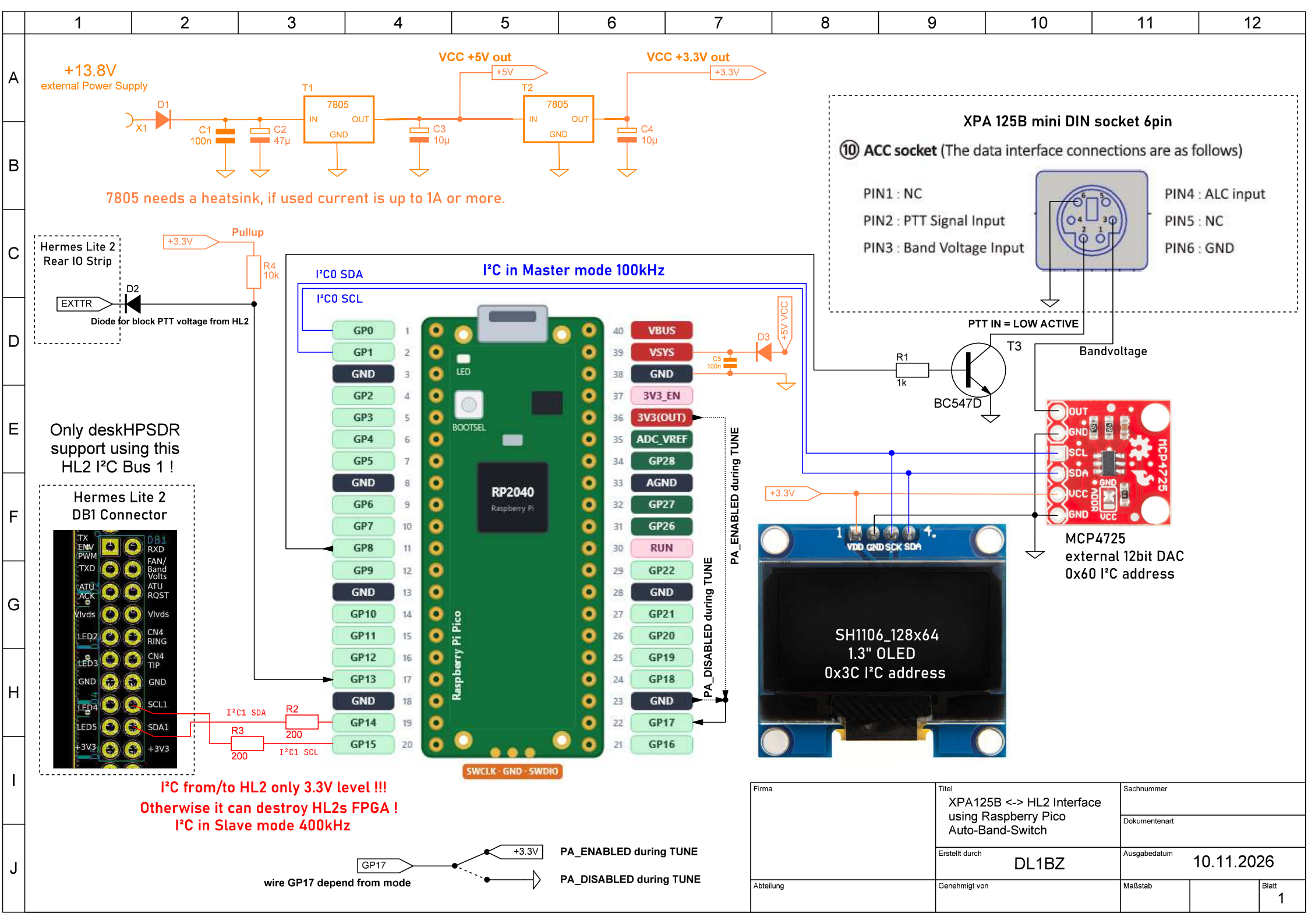

+13.8V
external Power Supply
D1
X1
C1
100n
C2
47µ
T1
7805
IN
OUT
GND
VCC +5V out
+5V
C3
10µ
T2
7805
IN
OUT
GND
VCC +3.3V out
+3.3V
C4
10µ
7805 needs a heatsink, if used current is up to 1A or more.
XPA 125B mini DIN socket 6pin
⑩ ACC socket (The data interface connections are as follows)
PIN1 : NC
PIN2 : PTT Signal Input
PIN3 : Band Voltage Input
PIN4 : ALC input
PIN5 : NC
PIN6 : GND
Hermes Lite 2
Rear IO Strip
EXTTR
D2
Diode for block PTT voltage from HL2
+3.3V
Pullup
R4
10k
I²C in Master mode 100kHz
I²C0 SDA
I²C0 SCL
PTT IN = LOW ACTIVE
T3
Bandvoltage
R1
1k
BC547D
D3
+5V VCC
C5
100n
GP0
GP1
GND
GP2
GP3
GP4
GP5
GND
GP6
GP7
GP8
GP9
GND
GP10
GP11
GP12
GP13
GND
GP14
GP15
LED
BOOTSEL
RP2040
Raspberry Pi
Raspberry Pi Pico
VBUS
VSYS
GND
3V3_EN
3V3(OUT)
ADC_VREF
GP28
AGND
GP27
GP26
RUN
GP22
GND
GP21
GP20
GP19
GP18
GND
GP17
GP16
SWCLK - GND - SWDIO
PA_ENABLED during TUNE
PA_DISABLED during TUNE
Only deskHPSDR
support using this
HL2 I²C Bus 1 !
Hermes Lite 2
DB1 Connector
DB1
TX
ENV
PWM
TXD
ATU
ACK
Vlvds
LED2
LED3
GND
LED4
LED5
+3V3
RXD
FAN/
Band
Volts
ATU
RQST
Vlvds
CN4
RING
CN4
TIP
GND
SCL1
SDA1
+3V3
I²C1 SDA
I²C1 SCL
R2
200
R3
200
+3.3V
OUT
GND
SCL
SDA
VCC
ADDR
GND
VCC
MCP4725
MCP4725
external 12bit DAC
0x60 I²C address
+3.3V
VDD GND SCK SDA
SH1106_128x64
1.3" OLED
0x3C I²C address
I²C from/to HL2 only 3.3V level !!!
Otherwise it can destroy HL2s FPGA !
I²C in Slave mode 400kHz
GP17
+3.3V
PA_ENABLED during TUNE
PA_DISABLED during TUNE
wire GP17 depend from mode
Firma
Titel
XPA125B <-> HL2 Interface
using Raspberry Pico
Auto-Band-Switch
Sachnummer
Dokumentenart
Erstellt durch
DL1BZ
Ausgabedatum
10.11.2026
Abteilung
Genehmigt von
Maßstab
Blatt
1